SAMRUDDHI / LDFRS

Leaf Disease Detection & Fertilizer Recommendation System

Comprehensive System Functionalities & Code Directory Map | Official Architecture Guide

1. Platform Overview & Architectural Stack

LDFRS (Samruddhi) is an end-to-end precision agriculture intelligence platform designed to empower farmers with instant leaf disease diagnosis, real scientific fertilizer prescriptions, proactive meteorological spray scheduling, crop recovery time-lapses, nearby agri-cooperative maps, and agronomist expert second-opinions. The architecture is engineered to run seamlessly both online and 100% offline in rural field conditions.

Three-Tier Architectural Stack:

- Frontend: Vite + React 19, React-Leaflet, React Router, Socket.io-client, CSS Custom Properties (Natural theme)
- Backend: Node.js (v24), Express 5, MongoDB / Mongoose, JWT, PDFKit, Socket.io, Axios, Multer
- ML Service: Python 3.11, FastAPI, PyTorch 2.14, Torchvision, PlantVillage 38-class Convolutional Neural Network

2.1 AI Leaf Disease Detection & Uncertainty Gating

Purpose: Deep learning computer vision system that classifies crop leaf diseases from photos, computes prediction confidence, and gates uncertainty.

Key Capabilities:

- Identifies 38 distinct crop disease classes across Tomato, Potato, Corn, Grape, Apple, Rice, Pepper, Peach, etc.
- Computes Softmax prediction confidence and Shannon entropy to detect model uncertainty.
- Automated Inference Gating: If confidence is < 70% or entropy is high, the scan is auto-enqueued to the Agronomist Expert Queue for second-opinion.
- Dual-mode inference: Prefers local offline FastAPI service (port 8001); falls back to cloud Render service or offline diagnostic engine.

Source Code Directories & File Paths:

Frontend UI	`frontend/src/components/ImageUpload.jsx`
Frontend View	`frontend/src/components/PredictionResult.jsx`
Controller	`backend/src/controllers/predictionController.js`
Inference Gate	`backend/src/services/inferenceGate.js`
ML Service Adapter	`backend/src/services/mlService.js`
ML Config & Failover	`backend/src/config/mlConfig.js`
Model Definition	`backend/src/models/Prediction.js`
FastAPI ML Service	`ml-service/app.py`
PyTorch Inference Engine	`ml-service/services/disease_prediction.py`
Trained Weights File	`ml-service/models/disease_model.pth (16.5 MB)`

2.2 Real-World Agronomic Fertilizer Prescriptions

Purpose: Replaces generic disclaimers with certified, commercial agronomic fertilizer formulations, dosages, methods, and precautions specific to the detected pathogen.

Key Capabilities:

- Tailored Formulations: Recommends real products like Potassium Nitrate (13-0-45), SOP (0-0-50), MKP (0-52-34), Calcium Nitrate with Boron, and Zinc EDTA.
- Precise Dosages: Gives actionable per-liter and per-acre measures (e.g. 5g/L foliar spray; 5 kg/acre drip fertigation).
- Actionable Application Methods: Detailed leaf canopy spraying instructions and root zone drenching.
- Agronomic Scientific Reasons: Explains the biochemical recovery mechanism (e.g., Potassium thickens epidermal cell walls and stimulates phytoalexins).
- Nutritional Precautions: High-priority warnings (e.g., WITHHOLD high-nitrogen Urea during active blight to prevent succulent spore-incubating tissue).

Source Code Directories & File Paths:

Prescription Engine	`backend/src/services/fertilizerRecommendationService.js`
Disease Info Service	`backend/src/services/diseaseInfoService.js`
Disease Knowledge Base	`backend/src/data/diseaseInformation.js`
Frontend Result Card	`frontend/src/components/PredictionResult.jsx`
PDF Report Sync	`backend/src/services/reportService.js`

2.3 Smart Spray Window & Farm Weather Advisor

Purpose: Real-time meteorological spray decision engine that prevents chemical wash-off, spray drift, and foliage scorch using WeatherAPI satellite forecast data.

Key Capabilities:

• Live Suitability Index: Color-coded status (Optimal / Caution / Do Not Spray) based on real temperature, wind speed, gusts, and rain probability.

• Automated Best Window Calculation: Scans next 24 hours of hourly conditions and outputs the exact safest slot (e.g., Today 6:30 AM – 9:30 AM).

• 4 Environmental Safety Factors: Rain wash-off risk (mm), Wind drift speed (km/h), Temperature foliage uptake (°C), and Leaf drying humidity (%).

• 10-Hour Visual Timeline: Horizontal strip displaying upcoming hours with weather condition icons, temperatures, rain chances, and safety status dots.

• Offline Agronomic Baseline: Provides field-proven safe spray guidelines when no internet connection is available.

Source Code Directories & File Paths:

Weather Forecast Service	`backend/src/services/weatherService.js`
Weather Risk Controller	`backend/src/controllers/weatherRiskController.js`
Weather Risk Routes	`backend/src/routes/weatherRiskRoutes.js`
Disease Risk Rules	`backend/src/data/diseaseWeatherRiskRules.js`
Frontend Spray Panel	`frontend/src/components/WeatherRiskPanel.jsx`
Dashboard Integration	`frontend/src/pages/Dashboard.jsx`

2.4 Nearby Fertilizer Shops with Interactive Red Pin Map

Purpose: Maps genuine regional fertilizer suppliers, Primary Agricultural Cooperative Societies (PACS), and Rythu Bharosa Kendras (RBK) around the farm on Leaflet.

Key Capabilities:

• Interactive Leaflet Map: Completely client-side rendered with custom red SVG pins for shops and blue pulse pin for the farmer location.

• Geodesic Projection: Places shops at authentic offsets (0.8 km, 1.6 km, etc.) from the farmer's GPS coordinates.

• Interactive Pin Popups: Shows store name, distance, address, one-tap Call button, WhatsApp order link, and Google Maps Navigation link.

• Card & Map Sync: Clicking "Pin on Map" on any shop card smoothly flies the camera to that exact red marker and opens its popup.

• Reverse Geocoding: Automatically resolves the farmer's town, mandal, district, and state via BigDataCloud & WeatherAPI.

Source Code Directories & File Paths:

Shop Discovery Service	`backend/src/services/shopService.js`
Shop Controller	`backend/src/controllers/shopController.js`
Shop Routes	`backend/src/routes/shopRoutes.js`
Frontend Map Component	`frontend/src/components/NearbyShops.jsx`
Dashboard Geolocation	`frontend/src/pages/Dashboard.jsx`

2.5 Crop Recovery Tracker & Time-Lapse Healing

Purpose: Tracks crop recovery progression over 7, 14, and 21 days with photo checkpoints, quantitative healing scores, and visual time-lapses.

Key Capabilities:

• Direct link from diagnosis into recovery journey with one click (Track Healing & Follow-up).

• Multi-stage checkpoints: Upload follow-up leaf images as treatment progresses.

• Side-by-side comparison slider: Visual comparison of original diseased leaf versus healed leaf.

• Quantitative progress metrics: Status badges (Recovering, Stabilized, Healed), recovery percentage bars, and expert verification.

Source Code Directories & File Paths:

Database Model	`backend/src/models/RecoveryTracker.js`
Controller	`backend/src/controllers/recoveryController.js`
Routes	`backend/src/routes/recoveryRoutes.js`
Frontend Page	`frontend/src/pages/RecoveryTracker.jsx`
App Navigation Route	`frontend/src/App.jsx (/recovery)`

2.6 Agronomist Expert Review System & Annotation Canvas

Purpose: Specialized workflow for agricultural researchers and extension officers to inspect, annotate, and verify low-confidence farmer scans.

Key Capabilities:

• Molecular Glassmorphic Theme: Distinct laboratory visual styling for agronomists.
• Triage Review Queue: Displays unconfirmed, high-entropy, or low-confidence farmer submissions.
• Interactive Annotation Canvas: Full-screen pan, zoom, and drawing tools allowing experts to draw bounding boxes and annotate fungal lesions directly on the leaf.
• Diagnosis Override & Advice: Experts update diagnosis, customize fertilizer dosage, and add tailored field instructions.
• Standalone MongoDB Sequential Fallback: Custom sequential atomic update mechanism eliminating replica set transaction errors on local machines.

Source Code Directories & File Paths:

Review Controller	`backend/src/controllers/reviewController.js`
Review Routes	`backend/src/routes/reviewRoutes.js`
Diagnosis Review Model	`backend/src/models/DiagnosisReview.js`
Expert Queue Page	`frontend/src/pages/ExpertQueue.jsx`
Expert Review Canvas	`frontend/src/pages/ExpertReview.jsx`
Farmer Status Badge	`frontend/src/components/DiagnosisReviewStatus.jsx`

2.7 Official PDF Advisory Report Generation & Delivery

Purpose: Generates professional, printable diagnostic reports and dispatches them via WhatsApp, Telegram, and Email.

Key Capabilities:

• PDFKit Engine: Generates branded PDF reports containing farmer details, scan timestamp, high-res leaf image, disease severity, fertilizer formulation, dosage, and precautions.
• Twilio WhatsApp Dispatch: Sends automated diagnosis summaries and download links directly to the farmer's WhatsApp phone number.
• Telegram Bot Alerts: Webhook-enabled real-time notification dispatch via Telegram.
• Direct Email Dispatch: Sends reports as PDF attachments via SMTP.

Source Code Directories & File Paths:

PDF Report Generator	`backend/src/services/reportService.js`
Delivery Controller	`backend/src/controllers/notificationController.js`
Notification Service	`backend/src/services/notificationService.js`
Delivery Modal UI	`frontend/src/components/ReportDeliveryDialog.jsx`

2.8 Community Disease Outbreak Map & Scan History

Purpose: Aggregates anonymized regional diagnosis reports across districts to alert communities of spreading crop epidemics.

Key Capabilities:

• Regional Disease Heatmap: Visualizes active outbreaks by crop type and severity across geographic districts.
• Farmer Scan Archive: Searchable historical list of all past diagnoses with timestamps, crop tags, and direct PDF downloads.
• Privacy-Preserving Geolocation: Coordinates are rounded and anonymized to protect exact farm boundaries.

Source Code Directories & File Paths:

History & Map Controller	`backend/src/controllers/predictionController.js`
Frontend History Page	`frontend/src/pages/History.jsx`
Community Leaflet Map	`frontend/src/components/ScanMap.jsx`

2.9 Government Agricultural Subsidies Finder

Purpose: Curated directory connecting farmers with central and state agricultural welfare schemes, subsidized fertilizers, and equipment grants.

Key Capabilities:

• Searchable Scheme Database: Covers PM-KISAN, Soil Health Card Scheme, Sub-Mission on Agricultural Mechanization (SMAM), and Micro-Irrigation grants.

• Direct Application Links: Official state and national portal URLs with clear eligibility criteria.

• Subsidized Fertilizer Quotas: Guidance on availing subsidized Neem-Coated Urea and Nano-DAP allocations.

Source Code Directories & File Paths:

Frontend Subsidies Page	`frontend/src/pages/Subsidies.jsx`
App Navigation Route	`frontend/src/App.jsx (/subsidies)`

2.10 Multi-Language Accessibility (Indic / Bhashini)

Purpose: Delivers full localization in regional Indian languages so rural farmers can navigate without language barriers.

Key Capabilities:

• Supported Languages: English, Hindi, Telugu, Tamil, Kannada, Marathi, Bengali.

• Dual Translation Engine: Integrates Bhashini pipeline URL with Google Translate & MyMemory fallback.

• Offline Graceful Fallback: If translation servers are unreachable, returns original text without crashing or throwing 502 errors.

Source Code Directories & File Paths:

Translation Service	`backend/src/services/translationService.js`
Translation Controller	`backend/src/controllers/translationController.js`
Translation Routes	`backend/src/routes/translationRoutes.js`
Language Selector UI	`frontend/src/components/LanguageSelector.jsx`

2.11 Security, Authentication & Role-Based Access Control

Purpose: Secures platform resources and ensures role-appropriate workflows for Farmers, Agronomist Experts, and Admins.

Key Capabilities:

• Role-Based Authorization: Strict route guards for Farmer, Expert, and Admin accounts.

• Bcrypt Password Hashing & JWT: Secure salted password hashing and stateless token issuance.

• Google OAuth Integration: One-tap authentication via Google Identity Services.

• Persistent Offline Sessions: Cached in localStorage; network disconnects never wipe user sessions.

Source Code Directories & File Paths:

User Model	`backend/src/models/User.js`
Auth Controller	`backend/src/controllers/authController.js`
Auth Routes	`backend/src/routes/authRoutes.js`
JWT Middleware Guard	`backend/src/middleware/authMiddleware.js`
Frontend Auth Context	`frontend/src/context/AuthContext.jsx`
Login / Register Pages	`frontend/src/pages/Login.jsx, Register.jsx`

2.12 UI Architecture, Dual Themes & Symmetric Layout

Purpose: Provides responsive, accessible, farmer-friendly design with specialized thematic modes.

Key Capabilities:

• Natural Agricultural Theme: Soft greens, earth tones, high-contrast readable typography, and crop emojis.

• Symmetric 50/50 Desktop Grid: Balanced 1:1 columns (repeat(2, minmax(0, 1fr))) and matching heights (min-height: 540px).

• Step-by-Step Stepper Banner: Visual progress tracker (1. Upload -> 2. AI Diagnosis -> 3. Nearby Shops on Maps).

• Expert Laboratory Theme: Molecular frosted glass overlay with backdrop-filter blur for researchers.

Source Code Directories & File Paths:

Master Stylesheet	`frontend/src/styles.css`
Main Dashboard Page	`frontend/src/pages/Dashboard.jsx`
App Shell & Navigation	`frontend/src/components/Navbar.jsx, Sidebar.jsx`

3. Repository Directory Structure Summary

`frontend/src/components/`	Reusable UI widgets: ImageUpload, PredictionResult, NearbyShops, WeatherRiskPanel, ScanMap
`frontend/src/pages/`	Route views: Dashboard, RecoveryTracker, ExpertQueue, ExpertReview, History, Subsidies, Login
`frontend/src/context/`	Global reactive state: AuthContext (persistent offline session management)
`frontend/src/styles.css`	Complete master CSS: natural agricultural theme, molecular expert theme, responsive grid
`backend/src/controllers/`	Express business logic: prediction, recovery, review, shop, weatherRisk, auth
`backend/src/services/`	Core engines: fertilizerRecommendation, weatherService, shopService, mlService, reportService
`backend/src/models/`	MongoDB schemas: User, Prediction, RecoveryTracker, DiagnosisReview
`backend/src/routes/`	REST API routing: predictionRoutes, recoveryRoutes, reviewRoutes, shopRoutes, weatherRiskRoutes
`backend/src/data/`	Agri knowledge base: diseaseInformation, diseaseWeatherRiskRules
`ml-service/`	FastAPI Python microservice, PyTorch model loader, disease_model.pth (38 classes)